

SOP-ENG-001

# Engineering Operations

## Medivac.ai Platform

Procedures for Licensed Aircraft Maintenance Engineers (LAMEs)  
and engineering staff — RFDS South Eastern Section

ROYAL FLYING DOCTOR SERVICE — SOUTH EASTERN SECTION

|              |                                                 |
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This procedure describes the standard engineering use of the Medivac.ai platform. It must be read together with the RFDS SE Maintenance Organisation Exposition (MOE), CASR Part 145, and approved maintenance data.

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# 1 Purpose and Scope

This Standard Operating Procedure (SOP) defines how Royal Flying Doctor Service South Eastern Section (RFDS SE) Licensed Aircraft Maintenance Engineers (LAMEs) and engineering support staff use the Medivac.ai platform to manage aircraft maintenance, defect rectification, engineering work orders, and airworthiness documentation across the RFDS SE fleet.

Medivac.ai is the authoritative digital system of record for the technical and journey log, engineering work orders, the maintenance planner, aircraft serviceability status, and asset utilisation. All engineering activity that affects the airworthiness of an aircraft must be recorded in Medivac.ai in accordance with this SOP and the RFDS SE Maintenance Organisation Exposition (MOE).

## Scope

- Applies to all RFDS SE LAMEs, unlicensed engineers, apprentices working under supervision, and maintenance planning staff at all engineering bases.
- Covers defect assessment, rectification, release to service, work order lifecycle, scheduled maintenance planning, aircraft status control, and emergency airworthiness actions performed within Medivac.ai.
- Does not replace the MOE, aircraft maintenance manuals (AMM), CASA regulatory instruments, or manufacturer instructions for continued airworthiness (ICA) — it governs how those obligations are recorded and tracked in the platform.

# 2 Fleet Reference

The following fixed-wing and rotary-wing types and engineering bases are configured in Medivac.ai. Aircraft registrations, serial numbers, and current maintenance status are maintained in the asset register; the reference below establishes the standard type and base nomenclature used throughout this SOP.

| Aircraft type              | Category                      | Typical role                      |
|----------------------------|-------------------------------|-----------------------------------|
| Beechcraft King Air B200   | Fixed wing — turboprop        | Aeromedical transfer / PTS        |
| Beechcraft King Air B350   | Fixed wing — turboprop        | Aeromedical / long-range transfer |
| Pilatus PC-12              | Fixed wing — single turboprop | Primary aeromedical retrieval     |
| Pilatus PC-24              | Fixed wing — jet              | High-speed critical retrieval     |
| Bombardier Challenger CL60 | Fixed wing — jet              | Long-range / high-acuity transfer |
| Helicopter (rotary wing)   | Rotary wing                   | Short-sector / access retrieval   |

## Engineering bases

| Base        | ICAO | Location                       |
|-------------|------|--------------------------------|
| Dubbo       | YSDU | New South Wales — central west |
| Broken Hill | YBHI | New South Wales — far west     |
| Bankstown   | YSBK | New South Wales — Sydney basin |
| Essendon    | YMEN | Victoria — Melbourne           |
| Launceston  | YMLT | Tasmania — northern            |

## 3 Tech & Journey Log

The Tech & Journey Log module in Medivac.ai is the electronic technical log for each aircraft. Engineers use it to review pilot-entered defects, assess and categorise them, document rectification, and record the release to service.

### 3.1 Reviewing pilot-entered defects

When a pilot raises a defect in the electronic tech log at the conclusion of a sector, Medivac.ai automatically generates a notification to the responsible engineering base. The duty LAME or maintenance controller is responsible for triaging new defects promptly.

- Notifications are routed to the base assigned to the aircraft and appear in the engineering dashboard defect queue.
- Each defect record shows the aircraft, sector, reporting pilot, date/time, defect description, and any pilot-applied deferral reference.
- Acknowledge each new defect to confirm it has entered the engineering workflow; acknowledgement is time-stamped against the engineer's Medivac.ai identity.

### 3.2 Defect assessment

The assessing LAME determines the airworthiness impact of each defect and records the categorisation in Medivac.ai. Categorisation drives whether the aircraft may continue in service and the applicable rectification interval.

| Category                     | Determination                                                                                                        | Effect on operations                                                                                 |
|------------------------------|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| MEL item                     | Defect permissible under the approved Minimum Equipment List with applicable (O)/(M) procedures and repair interval. | Aircraft may be released with the item deferred; rectification tracked to the MEL category interval. |
| Airworthiness-affecting      | Defect renders the aircraft unairworthy or falls outside MEL relief.                                                 | Aircraft must not be released until rectified; place Unserviceable per Section 6.2.                  |
| Non-airworthiness / cosmetic | No effect on airworthiness and MEL relief not required.                                                              | May be deferred to a convenient maintenance opportunity.                                             |

### 3.3 Rectification

On completing rectification, the engineer documents the work performed against the defect record. The entry must be sufficient to establish what was done, the data used, and the parts fitted.

- Record a clear description of the work performed and the approved maintenance data reference (AMM/CMM task, SB, or AD number).
- Log all parts used — part number, serial number (where applicable), batch/lot, and certification of conformity/airworthiness (Form 1 or equivalent).
- Attach supporting evidence (photographs, test results, torque values) where required by the task or the MOE.
- Confirm all tooling and independent inspections (duplicate/critical) required by the task have been completed and recorded.

### 3.4 CASA Part 145 compliance for maintenance records

Maintenance records created in Medivac.ai must satisfy the record-keeping requirements of CASR Part 145 and the RFDS SE MOE. Every maintenance record must be traceable, legible, and attributable to the certifying person.

- Records must identify the aircraft, the maintenance performed, the data used, the date of completion, and the person certifying the work.
- Certifications may only be made by appropriately authorised personnel holding the relevant category and scope on their RFDS SE authorisation.
- Records must not be altered after certification; corrections are made by a new, attributable entry that preserves the original record.
- Retain records for the period required by CASR Part 145 and the MOE (see Section 9).

### 3.5 Signing off releases to service

A Certificate of Release to Service (CRS) is issued within Medivac.ai only when all maintenance is complete, all required inspections are signed, and the aircraft is fit for the intended operation. The CRS is applied under the certifying engineer's unique Medivac.ai credentials, which constitute the electronic signature.

## 4 Engineering Work Orders

The work order module manages planned and unplanned maintenance tasks from initiation to closure. Work orders provide the audit trail linking defects and scheduled tasks to the labour, parts, and certifications applied.

### 4.1 Creating work orders

Work orders may be raised from a defect, a scheduled maintenance requirement, an Airworthiness Directive, or a Service Bulletin. When raised from a defect or planner item, Medivac.ai pre-populates the aircraft, task reference, and applicable maintenance data.

- Select the source — defect rectification, scheduled task, AD/SB, or modification.
- Confirm the aircraft, ATA chapter, task reference, and estimated labour.
- Attach the applicable maintenance data and any required planning documents before release to the shop floor.

### 4.2 Assigning work orders by base

Work orders are assigned to the engineering base responsible for the aircraft and then allocated to individual engineers according to authorisation and availability.

| Field             | Purpose                                                                          |
|-------------------|----------------------------------------------------------------------------------|
| Assigned base     | Routes the work order to Dubbo, Broken Hill, Bankstown, Essendon, or Launceston. |
| Assigned engineer | Names the responsible LAME/engineer; must hold the required category and type.   |
| Priority          | AOG, urgent, routine, or deferred — drives the planner and dashboard ordering.   |
| Target completion | Planned date/time used for forward planning and aircraft availability.           |

### 4.3 Parts requisition integration

Work orders integrate with the parts and inventory function so that parts can be requisitioned, reserved, and consumed against the specific task, maintaining full traceability.

- Requisition parts directly from the work order; stock availability and location are shown at the point of request.
- Reserved parts are held against the work order; consumption is recorded with part and serial numbers and certification references.
- Shortfalls raise a procurement action and, where they prevent completion, are flagged against aircraft availability.

### 4.4 Work order closure and release

A work order is closed only when all tasks are complete, all parts are booked out, all required inspections are signed, and the associated release to service (or CRS) has been issued. Medivac.ai prevents closure while mandatory sign-offs remain outstanding.

## 5 Maintenance Planner

The maintenance planner provides forward visibility of scheduled maintenance, component life limits, and aircraft out-of-service events. Engineering and planning staff use it to protect airworthiness and coordinate aircraft availability with Operations.

### 5.1 Scheduled maintenance tracking

- Track periodic inspections including 100-hourly and annual/periodic checks against hours flown, cycles, and calendar time.
- Monitor component life limits — hard-time, on-condition, and life-limited parts — with remaining life shown in hours, cycles, or calendar as applicable.
- Planner thresholds trigger alerts as tasks approach due, allowing work orders to be raised ahead of expiry.

### 5.2 Forward planning — 30 / 60 / 90 day forecast

The forward forecast projects upcoming maintenance across the fleet so that workload, parts, and aircraft availability can be planned. Standard horizons are 30, 60, and 90 days.

| Horizon      | Planning use                                                                        |
|--------------|-------------------------------------------------------------------------------------|
| Next 30 days | Confirm work orders, parts, and labour; lock aircraft out-of-service windows.       |
| Next 60 days | Order long-lead parts; balance workload across bases; flag potential conflicts.     |
| Next 90 days | Strategic view of heavy checks and component changes; capacity and hangar planning. |

### 5.3 Aircraft out-of-service planning

Planned maintenance that removes an aircraft from availability must be coordinated with Operations through Medivac.ai. Engineering proposes out-of-service windows; Operations confirms against the tasking and roster so that aeromedical capability is maintained.

### 5.4 Component change tracking

When a component is changed, the fitment is recorded so that life tracking continues against the new component and traceability is preserved for both the removed and fitted items.

- Record part number and serial number of the removed and fitted components.
- Record fitment date, aircraft hours/cycles at fitment, and the source certification (Form 1 / release note).
- The planner recalculates remaining life and next-due for the fitted component automatically.

## 6 Aircraft Status

Aircraft serviceability status in Medivac.ai must accurately reflect the true airworthiness state of each aircraft at all times. Operations, crewing, and tasking decisions rely on this status.

### 6.1 Engineering responsibility for accurate status

Engineering owns the accuracy of the serviceability status. Status must be updated immediately when the airworthiness state changes — on identification of an airworthiness-affecting defect, on commencement of maintenance that grounds the aircraft, and on completion of the release to service.

### 6.2 Placing aircraft Unserviceable (U/S)

An aircraft is placed Unserviceable when an airworthiness-affecting defect exists or when maintenance renders it not fit for flight. The U/S action is recorded with a reason and generates notifications to Operations and affected stakeholders.

- Set status to Unserviceable and record the reason, associated defect/work order, and expected return-to-service estimate where known.
- Medivac.ai notifies Operations and the relevant base so tasking can be reallocated.
- Keep the return-to-service estimate current as rectification progresses.

### 6.3 Return to service

An aircraft may only be returned to Serviceable status after the LAME has issued the release to service for the outstanding maintenance. The status update is gated on the sign-off — Medivac.ai will not allow return to Serviceable until the required certification is recorded.

## 7 Asset Utilisation

Engineering provides the airworthiness and component-life inputs that underpin fleet utilisation reporting, supporting decisions on aircraft allocation, maintenance timing, and fleet planning.

### 7.1 Engineering input to utilisation reports

- Ensure hours flown, cycles, and maintenance downtime are accurately captured so utilisation reflects true availability.
- Reconcile scheduled and unscheduled downtime against the planner to explain availability variances.

### 7.2 Component life tracking and reporting

- Report remaining component life and forecast component changes that will affect availability within the planning horizons.
- Highlight components approaching life limits so procurement and scheduling can act ahead of expiry.



## 8 Ground Vehicles

Ground vehicles and ground support equipment (GSE) supporting engineering and aeromedical operations are managed in Medivac.ai alongside aircraft assets, ensuring servicing and defects are tracked with the same discipline.

### 8.1 Ground vehicle maintenance scheduling

- Schedule periodic servicing for ground vehicles and GSE against calendar and usage intervals.
- Planner alerts flag upcoming servicing so equipment remains available and compliant.

### 8.2 Defect reporting for ground support equipment

- Raise defects against ground vehicles and GSE; unserviceable equipment is flagged to prevent use until rectified.
- Document rectification and return-to-service in the same manner as aircraft defects, maintaining a full equipment history.

## 9 Compliance

RFDS SE conducts maintenance as an Approved Maintenance Organisation. All engineering activity recorded in Medivac.ai must meet the following obligations.

### CASR Part 145 — Approved Maintenance Organisation

- Maintenance is performed and certified only within the scope of the RFDS SE Part 145 approval and the MOE.
- Certifying personnel hold current authorisations of the correct category, type, and scope for the work certified.
- Approved maintenance data is used for all tasks, and all required inspections are completed and recorded.

### Engineering records retention

- Maintenance records, certifications, and component histories are retained for the periods required by CASR Part 145 and the MOE.
- Records held in Medivac.ai are protected against unauthorised alteration and remain retrievable for audit and airworthiness review.

### RFDS Engineering EBA 2025

- Rostering, hours of work, on-call and callout arrangements, and licensing/authorisation provisions are applied in accordance with the RFDS Engineering Enterprise Bargaining Agreement 2025.
- Where the EBA affects engineering availability (e.g. fatigue and callout limits), this is reflected in planning and assignment within Medivac.ai.

## 10 Emergency Airworthiness Actions

Emergency airworthiness situations demand immediate, disciplined action and accurate recording. The following procedures apply and are tracked within Medivac.ai.

### 10.1 AOG — Aircraft on Ground

An AOG is declared when an aircraft is grounded by a serious defect or unserviceability requiring urgent engineering action. Rapid notification protects aeromedical capability and mobilises the right resources.

| Step | Action                                                                                 | Responsible      |
|------|----------------------------------------------------------------------------------------|------------------|
| 1    | Confirm and record AOG status; set aircraft Unserviceable in Medivac.ai.               | Duty LAME        |
| 2    | Notify Operations, the base, and the on-call engineering lead immediately.             | Duty LAME / MC   |
| 3    | Raise an AOG-priority work order and initiate parts/logistics action.                  | Duty LAME        |
| 4    | Maintain return-to-service estimate; escalate to Chief Engineer if capability at risk. | Engineering lead |

### 10.2 Airworthiness Directive response

Airworthiness Directives (ADs) are mandatory. Each applicable AD is recorded in Medivac.ai, assessed for applicability and compliance time, and tracked to completion through a work order.

- Log the AD reference, applicability, and compliance deadline (calendar, hours, or cycles).
- Raise a work order to accomplish the AD and record the method of compliance and evidence.
- Track recurring AD requirements so subsequent compliance is scheduled by the planner.

### 10.3 Mandatory Occurrence Reports (MOR)

Reportable occurrences identified during maintenance must be reported in accordance with the Transport Safety Investigation regime and RFDS SE procedures, coordinated with the Safety Officer.

- Identify and record reportable matters (e.g. major defects, airworthiness concerns) arising from maintenance.
- Coordinate with the RFDS SE Safety Officer to ensure MOR obligations to the ATSB/CASA are met within required timeframes.
- Link the occurrence to the relevant defect, work order, and any resulting corrective action in Medivac.ai.

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End of SOP-ENG-001. This document is a controlled procedure of RFDS South Eastern Section. It is UNCONTROLLED WHEN PRINTED. Refer to Medivac.ai and the RFDS SE MOE for the current approved version.